

Problem 1

Let $\varepsilon > 0$. We want to show that there exists $N \in \mathbb{N}$ such that for all $n > N$,

$$\left| \frac{5n+2}{8n-3} - \frac{5}{8} \right| < \varepsilon.$$

Consider

$$\begin{aligned} \left| \frac{5n+2}{8n-3} - \frac{5}{8} \right| &= \left| \frac{8(5n+2) - 5(8n-3)}{8(8n-3)} \right| \\ &= \left| \frac{31}{8(8n-3)} \right|. \end{aligned}$$

Notice that for $n \geq 1$, the denominator is positive:

$$8(8n-3) > 0 \quad \text{and the numerator } 31 > 0.$$

Hence the fraction is positive, so the absolute value can be dropped:

$$\left| \frac{31}{8(8n-3)} \right| = \frac{31}{8(8n-3)}.$$

Furthermore, for $n \geq 1$ we have

$$8n-3 \geq 4n \implies \frac{31}{8(8n-3)} \leq \frac{31}{8 \cdot 4n} = \frac{31}{32n}.$$

Now choose $N \in \mathbb{N}$ such that

$$\frac{31}{32N} < \varepsilon,$$

that is,

$$N > \frac{31}{32\varepsilon}.$$

Let

$$N = \left\lceil \frac{31}{32\varepsilon} \right\rceil.$$

Then for all $n > N$,

$$\left| \frac{5n+2}{8n-3} - \frac{5}{8} \right| \leq \frac{31}{32n} < \varepsilon.$$

Therefore,

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall n > N, \left| \frac{5n+2}{8n-3} - \frac{5}{8} \right| < \varepsilon.$$

Problem 2

We want to compute

$$\lim_{n \rightarrow \infty} \left(\sqrt{4n^2 + 3n + 8} - 2n \right).$$

To simplify, multiply and divide by the conjugate:

$$\sqrt{4n^2 + 3n + 8} - 2n = \frac{(\sqrt{4n^2 + 3n + 8} - 2n)(\sqrt{4n^2 + 3n + 8} + 2n)}{\sqrt{4n^2 + 3n + 8} + 2n}.$$

The numerator simplifies using the difference of squares:

$$(\sqrt{4n^2 + 3n + 8})^2 - (2n)^2 = (4n^2 + 3n + 8) - 4n^2 = 3n + 8.$$

So we have

$$\sqrt{4n^2 + 3n + 8} - 2n = \frac{3n + 8}{\sqrt{4n^2 + 3n + 8} + 2n}.$$

Factor n from numerator and denominator:

$$\frac{3n + 8}{\sqrt{4n^2 + 3n + 8} + 2n} = \frac{n(3 + 8/n)}{n \left(\sqrt{4 + 3/n + 8/n^2} + 2 \right)} = \frac{3 + 8/n}{\sqrt{4 + 3/n + 8/n^2} + 2}.$$

Now take the limit as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} \frac{3 + 8/n}{\sqrt{4 + 3/n + 8/n^2} + 2} = \frac{3 + 0}{\sqrt{4 + 0 + 0} + 2} = \frac{3}{2 + 2} = \frac{3}{4}.$$

Hence,

$$\lim_{n \rightarrow \infty} \left(\sqrt{4n^2 + 3n + 8} - 2n \right) = \frac{3}{4}.$$

Problem 3

(a)

Let $\varepsilon > 0$. We want to show that there exists $N \in \mathbb{N}$ such that for all $n > N$,

$$\left| \frac{3n^2 - n}{6n^2 + 1} - \frac{1}{2} \right| < \varepsilon.$$

Consider

$$\begin{aligned} \left| \frac{3n^2 - n}{6n^2 + 1} - \frac{1}{2} \right| &= \left| \frac{2(3n^2 - n) - (6n^2 + 1)}{2(6n^2 + 1)} \right| \\ &= \left| \frac{-2n - 1}{2(6n^2 + 1)} \right| \\ &= \frac{2n + 1}{2(6n^2 + 1)}. \end{aligned}$$

Notice that for $n \geq 1$, the denominator is positive:

$$2(6n^2 + 1) > 0 \quad \text{and the numerator } 2n + 1 > 0.$$

Hence the fraction is positive, so the absolute value can be dropped:

$$\left| \frac{-2n - 1}{2(6n^2 + 1)} \right| = \frac{2n + 1}{2(6n^2 + 1)}.$$

Furthermore, for $n \geq 1$ we have

$$2n + 1 \leq 3n \quad \text{and} \quad 2(6n^2 + 1) \geq 12n^2,$$

so

$$\frac{2n + 1}{2(6n^2 + 1)} \leq \frac{3n}{12n^2} = \frac{1}{4n}.$$

Now, choose $N \in \mathbb{N}$ such that

$$\frac{1}{4N} < \varepsilon,$$

that is,

$$N > \frac{1}{4\varepsilon}.$$

Let

$$N = \left\lceil \frac{1}{4\varepsilon} \right\rceil.$$

Then for all $n > N$,

$$\left| \frac{3n^2 - n}{6n^2 + 1} - \frac{1}{2} \right| \leq \frac{1}{4n} < \varepsilon.$$

Therefore,

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall n > N, \left| \frac{3n^2 - n}{6n^2 + 1} - \frac{1}{2} \right| < \varepsilon.$$

(b)

Suppose, for the sake of contradiction, that the sequence converges to some limit $x \in \mathbb{R}$. By the definition of the limit, for any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n > N$,

$$|x_n - x| < \varepsilon.$$

Take $\varepsilon = 1$. Then there exists $N \in \mathbb{N}$ such that for all $n > N$,

$$\left| \sqrt{2n} - 1 - x \right| < 1.$$

This inequality implies

$$-1 < \sqrt{2n} - 1 - x < 1 \quad \implies \quad x < \sqrt{2n} < x + 2.$$

Squaring the upper bound, we get

$$2n < (x + 2)^2 \quad \implies \quad n < \frac{(x + 2)^2}{2}.$$

However, $n \in \mathbb{N}$ can be arbitrarily large, so this inequality cannot hold for all n . This is a contradiction. Therefore, the sequence (x_n) does **not converge**.

Problem 4

Let (x_n) be defined by

$$x_{n+1} = \frac{3}{5}x_n + \frac{7}{5}, \quad x_1 = 2.$$

(a)

Claim: For all $n \in \mathbb{N}$,

$$x_{n+1} - x_n \geq 0 \quad \text{and} \quad x_n \leq \frac{7}{2}.$$

Proof (by induction):

Base case: For $n = 1$, we have

$$x_1 = 2 \leq \frac{7}{2}, \quad x_2 = \frac{3}{5}x_1 + \frac{7}{5} = \frac{6}{5} + \frac{7}{5} = \frac{13}{5} = 2.6 \geq x_1.$$

Hence the claim holds for $n = 1$.

Inductive step: Assume that for some $n \in \mathbb{N}$,

$$x_n \leq \frac{7}{2} \quad \text{and} \quad x_n - x_{n-1} \geq 0.$$

Then

$$x_{n+1} = \frac{3}{5}x_n + \frac{7}{5} \leq \frac{3}{5} \cdot \frac{7}{2} + \frac{7}{5} = \frac{21}{10} + \frac{14}{10} = \frac{35}{10} = \frac{7}{2}.$$

Hence $x_{n+1} \leq \frac{7}{2}$.

Next, compute

$$x_{n+1} - x_n = \frac{3}{5}x_n + \frac{7}{5} - x_n = \frac{3}{5}x_n - \frac{5}{5}x_n + \frac{7}{5} = \frac{7 - 2x_n}{5}.$$

Since $x_n \leq \frac{7}{2}$, we have

$$7 - 2x_n \geq 0 \quad \implies \quad x_{n+1} - x_n \geq 0.$$

Therefore, by the principle of mathematical induction, for all $n \in \mathbb{N}$,

$$x_n \leq \frac{7}{2} \quad \text{and} \quad x_{n+1} - x_n \geq 0.$$

□

(b)

From part (a), (x_n) is increasing and bounded above by $\frac{7}{2}$.

By the Bounded Monotone Convergence Theorem, every monotone increasing sequence that is bounded above converges. Therefore, (x_n) converges. Let

$$x = \lim_{n \rightarrow \infty} x_n.$$

Taking the limit in the recursive formula

$$x_{n+1} = \frac{3}{5}x_n + \frac{7}{5},$$

and using the properties of limits, we get

$$x = \frac{3}{5}x + \frac{7}{5}.$$

Solving for x :

$$x = \frac{3}{5}x + \frac{7}{5}$$

$$5x = 3x + 7$$

$$2x = 7$$

$$x = \frac{7}{2}.$$

Thus,

$$\lim_{n \rightarrow \infty} x_n = \frac{7}{2}.$$

Problem 5

$$\text{a) } x_n = \left(\frac{4}{17}\right)^n, \quad y_n = \left(\frac{1}{2}\right)^n, \quad \frac{x_n}{y_n} \rightarrow 0$$

$$\text{b) } x_n = \frac{19}{7n+13}, \quad y_n = \frac{3n-5}{4n^2+13}, \quad \frac{x_n}{y_n} \rightarrow \frac{76}{21} > 0$$

$$\text{c) } x_n = \frac{1}{n}, \quad y_n = \frac{1}{n^2}, \quad \frac{x_n}{y_n} = n \text{ does not converge}$$